(i) stdio.h

(ii) stdlib.h

(iii) conio.h

(iv) iostream.h

(c)
```
#include<stdio.h>

{

char ch = ‘Z’

printf (“% d\n”,ch);

return 0;

}
```

(i) 65 (ii) 90 (iii) 97 (iv)122

(d) How is an array initialized in C language?

(i) int a [3] = {1,2,3};

(ii) int a = {1,2,3};

(iii) int a[ ] = new int [3];

(iv) int a(3) = [1,2,3];

(e) What is the return type of the fopen() function in C?

(i) Pointer to a FILE object.

(ii) Pointer to an integer.

(iii) An integer.

(iv) None of the above

(f) What will be the output of the following C Code?

```
#include<stdio.h>

Int main()

{

Int x = 4, y, z;
```

Y = --x;

Z = x --;

printf(“%d%d%d”, x,y,z);

(i) 3 2 3

(ii) 2 2 3

(h) #include<stdio.h>

Int main ( )

{

float a = 5, b = 2;

int c,d;

c = a/b;

d = c/2;

printf(“%”, d)’

retrun 0;

}

(i) 1 (ii) 0 (iii) 1.5 (iv) 1.25

(j) In C, if you pass an array as an argument to a function, what actually gets passed?

(i) First element of the array

(ii)Value of elements of the array

(iii) Base address of the array

(iv) Address of the last element of array

(i) How is the 3^{nd} element in an array accessed based on pointer notation

(i) *a+3

(ii) *(a+3)

(iii) *(*a+3)

(iv) &(a+3)

Q.2 (a) Explain entry controlled loop and exit controlled loop with flow charts and examples.

(b) Write a C program to reverse a given multi-digit number.

Q.3 (a) Explain break and continue keywords with suitable examples in context of managing loops.

(b) What is storage classes in C. Write features (storage, default value, scope, life) of variables defined under each storage class.

Q.4 (a) Write a C function for Bubble sort. Analyse the time complexity of Bubble sort for each standard cases.

(b) Differentiate between formal argument and actual argument with an example.

Q.5 (a) What is recursion? Write a recursive C program to generate n" term of the Fibonacci series.

(b) How string is declared and initialized? Explain any four predefined string manipulation functions with examples. https://www.akubihar.com

Q. 6 (a) write a C program to find result matrix after multiplying two given matrices using 2-D arrays.

(b) Write a program to copy contents of one file to another. While doing so replace all lowercase characters to their equivalent uppercase characters.

Q.7 (a) Write the difference between structure and union. Compare them with the help of an example.

(b) Write a program to copy the contents of one array into another in reverse order.

Q.8 (a) Differentiate between call-by-value and call-by-reference with suitable examples.

(b) Discuss conditional operator? Write a C program to find largest of three numbers using conditional operator?

Q.9 (a) Write a C program that converts a string like “12” to an integer 124.

(b) Write a C program using the nesting of loops to print the following pattern:

Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Course: B.Tech. | **Time: 03 Hours**
Code: 100104 | ***Subject: Programming for Problem Solving*** | **Full Marks: 70**

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** [2 x 7 = 14]

(a) What is the output of the following code?

```
int x = 10;
x++;
printf("%d", x);
```

(i) Error (ii) 10 (iii) 11 (iv) 9

(b) Which function is used to dynamically allocate memory in C?

(i) malloc() (ii) calloc() (iii) realloc() (iv) all of the above

(c) What is the output of the following code?

```
#include<stdio.h>
int main() {
char *p = 0;
*p = 'a';
printf("%c", *p);
return 0;}
```

(i) It will print 'a' (ii) It will print 0 (iii) Compile time error (iv) Runtime error

(d) What are the elements present in the array of the following C code?

```
int array[6] = {5};
```

(i) 5, 5, 5, 5, 5, 5 (ii) 5, 0, 0, 0, 0, 0
(iii) 6, 6, 6, 6, 6, 6 (iv) (garbage), (garbage), (garbage), (garbage), (garbage), 5

(e) What will be the output of the following C code?

```
void main(){
int i = 0;
while (i < 10){
i++;
printf("hi\n");
while (i < 8){
i++;
printf("hello\n");}}}
```

(i) hi is printed 8 times, hello 7 times and then hi 2 times
(ii) hi is printed once, hello 7 times
(iii) hi is printed once, hello 7 times and then hi 2 times
(iv) None of the above.

(f) What does the following declaration mean?

*int (*ptr)* [10];

(i) ptr is array of pointers to 10 integers. (ii) ptr is a pointer to an array of 10 integers.
(iii) ptr is an array of 10 integers. (iv) ptr is a pointer to array of 10 characters.

(g) C is ________ type of programming language?

(i) Object Oriented (ii) Procedural
(iii) Bit level language (iv) Functional

(h) Choose the right C statement.

(i) int my_age = 10; (ii) int my.age = 10;
(iii) my age = 10; (iv) All are correct

(i) Consider the following integer 2D array, which is stored in memory with the base address 1400. What would be the address of the element 14? [Assume that an integer value takes 4 bytes of memory space.]

```
int a[3][5]={{1,2,3,4,5},{6,7,8,9,10},{11,12,13,14,15}};
```

(i) 1440 (ii) 1444 (iii) 1448 (iv) 1452

(j) Consider the following declaration of the variables:

```
int x, A[5][6], *p=&A[0][0];
```

Which of the following is/are the correct expression(s) to access A[3][5] and assign it to x?

(i) x=*(p+23); (ii) x=*(p+28); (iii) x=*(*(A+5)+3); (iv) x=*(*(A+3)+5);

Q.2 (a) Explain the difference between auto and static type storage variables with suitable example codes. [7]

(b) Differentiate between structures and unions in C. Write a C program to detect whether the computer is little endian or big endian using unions. [7]

Q.3 (a) Write an efficient C program to search if the integer 24 is present in the given sorted array or not. If it is present, print the index where it is found. [7]

(b) Explain the insertion sort algorithm and its steps using an example. Consider the array: [5, 2, 4, 6, 1, 3]. [7]

Q.4 (a) Describe the concept of pointers in C programming language. Explain how pointers work and discuss their importance in programming, particularly in memory management and efficient data manipulation. [7]

(b) Write a C program to copy contents of one file to another. While doing so replace all lowercase characters to their equivalent uppercase characters. [7]

Q.5 (a) Write a C Program to find the Euclidian distance between two points P(x, y) and Q(a, b) using structure representation of points P and Q. [7]

(b) Differentiate between call-by-value and call-by-reference with suitable examples. [7]

Q.6 (a) Define arrays in C and explain their declaration and usage. Write a C program to reverse the content of an integer array without using auxiliary array. [7]

(b) Define implicit and explicit type casting in C. Provide code examples illustrating when each type of casting occurs and their potential side effects. [7]

Q.7 (a) Explain the concept of nested loops. Describe how the **break** and **continue** statements alter the flow of control within C loops. [7]

(b) Explain the concept of algorithmic complexity and its significance in computer science. Using examples, illustrate how different algorithms exhibit various orders of complexity, such as constant time (**O(1)**), linear time (**O(n)**), logarithmic time (**O(log n)**), quadratic time (**O(n^2)**), and exponential time (**O(2^n)**). [7]

Q.8 (a) The first 6 numbers of the Fibonacci series are: 0 1 1 2 3 5. Write a C program using recursion to find and print the Fibonacci series for first *n* terms. [7]

(b) Write a C program using recursion to find factorial of a given number *n*. [7]

Q.9 (a) Write a C program for the following: [7]

(i) Reverse a string without using strrev().

(ii) Concatenate two strings without using strcat().

(b) Write a C program to find result matrix after multiplying two given matrices using 2-D arrays. Check the multiplication criteria also while deriving the logic. [7]

Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024

Course: B.Tech
Code: 100111

Subject: Programming for Problem Solving

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer the following (Any seven question only):*** **[2 × 7 = 14]**

(a) Which of the following is a valid variable name in C?
(i) 2value (ii) @value
(iii) value_2 (iv) value#2

(b) Which of these is not a valid data type in C?
(i) int (ii) real
(iii) float (iv) char

(c) Which of the following is the correct syntax for if statement?
(i) if [condition] then (ii) if condition then
(iii) if (condition) (iv) if {condition}

(d) What is the output of printf("%d", 5 + 3 * 2);?
(i) 16 (ii) 10
(iii) 11 (iv) 13

(e) Which header file is required to use printf()?
(i) stdlib.h (ii) stdio.h
(iii) conio.h (iv) string.h

(f) Which keyword is used to define a constant in C?
(i) final (ii) #define
(iii) constant (iv) static

(g) What is the index of the first element in an array?
(i) 1 (ii) -1
(iii) 0 (iv) Depends on array

(h) Which one is the logical AND operator in C?
(i) & (ii) |
(iii) && (iv) ||

(i) Which format specifier is used for printing float values?
(i) %d (ii) %c
(iii) %f (iv) %x

(j) Which of these is not a loop in C?
(i) for (ii) do-while
(iii) while (iv) repeat-until

Q.2 (a) Explain the structure of a C program with an example. **[7]**
(b) Describe basic data types in C with memory size. **[7]**

Q.3 (a) What is an algorithm. Write an algorithm to perform the addition of two numbers in C and draw a flowchart to represent the addition of two numbers. **[7]**
(b) Write a program to check whether a number is even or odd using if-else. **[7]**

Q.4	(a)	Explain Arithmetic and Logical operators in C with example programs.	[7]
	(b)	Explain Relational and Bitwise operators with example programs.	[7]
Q.5	(a)	Explain call by value and call by reference with examples.	[7]
	(b)	What is the difference between library functions and user-defined functions?	[7]
Q.6	(a)	What is an array in C programming? Write advantages of using arrays. Explain a one-dimensional array with a suitable example program.	[7]
	(b)	Describe the difference between character array and string.	[7]
Q.7	(a)	Write the difference Between Array and Structure in C programming.	[7]
	(b)	Write the difference Between Structure and Union in C programming.	[7]
Q.8	(a)	What is a pointer? Explain pointer declaration and initialization with sample program.	[7]
	(b)	What is file? Explain text file and binary file in details.	[7]

Q.9 Write short notes on ***any two*** of the following:- [7 x 2 = 14]

(a) Structure
(b) Union
(c) Recursion
(d) Fibonacci series program